(i.e) $\Delta y_0 = y_1 - y_0, \Delta y_1 = y_2 - y_1, \Delta y_2 = y_3 - y_2, \ldots, \Delta y_{n-1} = y_n - y_{n-1}$

In general, $\Delta y_n = y_{n+1} - y_n$, $n = 0,1,2,3,\ldots$

The symbol Δ is called the forward difference operator and pronounced as delta.

The forward difference operator Δ can also be defined as $\Delta f(x) = f(x+h) - f(x)$, h is the equal interval of spacing.

Proof of these properties are not included in our syllabus:

Properties of the operator Δ :

Property 1: If c is a constant then $\Delta c = 0$

Proof: Let $f(x) = c$

$\therefore f(x+h) = c$ (where 'h' is the interval of difference)

$$\Delta f(x) = f(x+h) - f(x)$$

$$\Delta c = c - c = 0$$

Property 2: Δ is distributive i.e. $\Delta(f(x) + g(x)) = \Delta f(x) + \Delta g(x)$

Proof: $\Delta[f(x) + g(x)]$

$$= [f(x+h) + g(x+h)] - [f(x) + g(x)]$$

$$= f(x+h) + g(x+h) - f(x) - g(x)$$

$$= f(x+h) - f(x) + g(x+h) - g(x)$$

$$= \Delta f(x) + \Delta g(x)$$

Similarly we can show that $\Delta[f(x) - g(x)] = \Delta f(x) - \Delta g(x)$

In general, $\Delta[f_1(x) + f_2(x) \ldots\ldots + f_n(x)] = \Delta f_1(x) + \Delta f_2(x) + \ldots + \Delta f_n(x)$

Property 3: If c is a constant then $\Delta c f(x) = c \Delta f(x)$

Proof: $\Delta[c f(x)] = c f(x+h) - c f(x)$

$$= c[f(x+h) - f(x)]$$

$$= c \Delta f(x)$$

Results without proof

1. If m and n are positive integers then $\Delta^m . \Delta^n f(x) = \Delta^{m+n} f(x)$
2. $\Delta[f(x)\, g(x)] = f(x)\Delta g(x) + g(x)\Delta f(x)$
3. $\Delta\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)\Delta f(x) - f(x)\Delta g(x)}{g(x).\ g(x+h).}$

The differences of the first differences denoted by $\Delta^2 y_0, \Delta^2 y_1, \ldots, \Delta^2 y_n$ are called second differences, where

$$\Delta^2 y_n = \Delta(\Delta y_n) = \Delta(y_{n+1} - y_n)$$

$$= \Delta y_{n+1} - \Delta y_n$$

$$\Delta^2 y_n = \Delta y_{n+1} - \Delta y_n,\ n = 0,1,2,\ldots$$

$$\Delta^2 y_0 = \Delta y_1 - \Delta y_0$$

$$\Delta^2 y_1 = \Delta y_2 - \Delta y_1$$

Similarly the differences of second differences are called third differences.

$$\Delta^3 y_n = \Delta^2 y_{n+1} - \Delta^2 y_n,\ n = 0,1,2,\ldots$$

In particular,

$$\Delta^3 y_0 = \Delta^2 y_1 - \Delta^2 y_0$$

$$\Delta^3 y_1 = \Delta^2 y_2 - \Delta^2 y_1$$

In general kth differences of yn is

$$\Delta^k y_n = \Delta^{k-1} y_{n+1} - \Delta^{k-1} y_n,\ n = 0,1,2,\ldots$$

Note

$$\Delta^k f(x) = \Delta^{k-1} f(x+h) - \Delta^{k-1} f(x)$$

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